HackPSU Project Brief: Insurance Denial Appeal Assistant

The Problem (30 seconds)

When health insurance denies a claim, most people just pay out of pocket — even though **over half of denials are overturned on appeal**. The problem isn't that appeals are hard to win. The problem is that people don't know they *can* appeal, don't understand *why* they were denied, and don't know *how* to write the appeal letter. Insurance companies profit from this confusion.

Under the ACA, **every insured person has the legal right to appeal without a lawyer** — first an internal appeal (insurer reviews its own decision), then an external appeal (independent third party, binding on the insurer). Our tool helps people actually exercise that right.

What We're Building

A web app where a user **uploads their denial letter or Explanation of Benefits (EOB)**, and the system:

1. **Extracts structured data** from the document (denial reason code, procedure code, plan type)
2. **Looks up the applicable regulation** from a curated legal database
3. **Generates a draft appeal letter** with proper regulatory citations

The pitch line: *"We don't promise to win your appeal. We promise that if you have grounds to appeal, you won't lose just because you didn't know how to ask."*

Why This Isn't Just "Paste It Into ChatGPT"

The key differentiator is a **three-stage pipeline** — the LLM is never doing freeform legal reasoning:

Stage 1 — Structured Extraction

The LLM reads the denial letter and extracts specific fields:

- **CARC/RARC codes** (standardized denial reason codes published by CMS — these are on every EOB)
- **CPT/HCPCS procedure codes** (what procedure was denied)
- **Stated reason for denial** (plain text)
- **Plan type** (ACA marketplace, employer-sponsored, etc.)

Stage 2 — Regulatory Lookup (RAG)

The extracted denial code is used to **query a curated database** of:

- ACA appeal regulations (42 USC §300gg-19)
- CMS denial reason code definitions and their regulatory counterparts
- Common appeal grounds mapped to each denial type
- Template appeal language that has historically worked

This is **deterministic lookup, not LLM interpretation.** Code CO-50 maps to specific regulations every time.

Stage 3 — Appeal Letter Generation

The LLM composes a draft appeal letter using:

- The extracted facts from Stage 1
- The specific regulations and appeal grounds from Stage 2
- Proper procedural language (deadlines, required formatting, escalation paths)

Every citation in the letter comes from the RAG database, not LLM generation. The user can verify each one.

Why This Architecture Matters for Judging

When judges ask "how do you know this is accurate?", the answer is:

- The denial codes are **standardized and published by CMS** — we're reading them, not interpreting them
- The regulations are **pulled from a curated database**, not hallucinated
- The LLM's role is **extraction and composition**, not legal reasoning
- If the system can't confidently identify the denial basis, **it says so** instead of guessing

Scope for 24 Hours

We're handling the **top 5-6 most common denial reasons** (covers the vast majority of consumer denials):

1. Prior authorization not obtained
2. Out-of-network provider
3. Medical necessity not demonstrated
4. Coding/billing error
5. Timely filing limit exceeded
6. Experimental/investigational treatment

We are **not** handling: ERISA-specific edge cases, Medicare/Medicaid (different regulatory framework), or multi-state jurisdictional complexity. We scope to ACA marketplace plans.

Tech Stack

Layer	Tool	Why
Frontend	React + Tailwind	Upload UI, display parsed denial + appeal letter side by side
Backend	Python / FastAPI	Orchestrates the three-stage pipeline
Extraction	LLM (Gemini Free Tier)	Structured field extraction from uploaded document
RAG Database	Vector store or simple JSON lookup	Curated ACA regulations, CARC/RARC mappings, appeal templates
Appeal Generation	LLM (Gemini Free Tier)	Composes letter from verified components
OCR (if needed)	Tesseract / pdf-to-text	For scanned/photographed denial letters

Demo Plan

We prepare **3 synthetic denial letters** (no real PHI needed):

1. **Prior Auth Denial** — Tool extracts code, finds the regulation requiring insurers to provide retroactive auth pathways, generates appeal letter citing it
2. **Medical Necessity Denial** — Tool identifies the denial basis, pulls the ACA requirement that denials include clinical rationale, generates appeal requesting peer-to-peer review
3. **Live Upload** — Judge or audience member uploads a sample, system processes it end-to-end

What We're NOT Claiming

- We're not giving legal advice — we're helping people use a government-mandated appeals process
- We're not guaranteeing appeal success — we're ensuring the appeal is properly structured
- We're not replacing lawyers — for complex ERISA cases, we explicitly recommend consulting one

Task Division (Suggested)

- **Person 1**: Data — build the RAG database (CARC/RARC mappings, ACA regulations, appeal templates)
- **Person 2**: Backend — FastAPI pipeline, LLM integration, extraction → lookup → generation flow
- **Person 3**: Frontend — upload interface, side-by-side display (denial breakdown | draft appeal letter)
- **Everyone**: Demo prep, synthetic denial letter creation, testing